

Problem-Solution Fit

Date	4 july 2026
Team ID	SWTID-2026-5594
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

☐ Credit Card Applicants ☐ Banks and Financial Institutions ☐ Credit Analysts ☐ Financial Compliance Officers	Before • Anxious • Uncertain • Frustrated After • Confident • Satisfied • Happy with faster results	Manual Verification • Pros: Human judgment, detailed review. • Cons: Time-consuming, inconsistent, prone to errors. Basic Rule-Based Systems • Pros: Simple to implement. • Cons: Low prediction accuracy and limited flexibility.
• Manual credit card approval takes a long time. • Human errors may occur during verification. • High volume of daily applications delays decisions. • Applicants are uncertain about their	• Manual verification process. • Large number of applications. • Human decision-making errors. • Lack of intelligent automation.	Develop an AI-powered Credit Card Approval Prediction System using Machine Learning. The system analyzes applicant details such as income, employment, education, family information, and credit-related features using a trained Random Forest model. It provides instant Approved/Rejected

eligibility. Frequency: High (Occurs daily)	7. BEHAVIOR + ITS INTENSITY [BE] • Users frequently check their eligibility before applying. • Banks process thousands of applications every day. Intensity: High	predictions through a Flask web application, reducing processing time, improving prediction accuracy, and supporting faster decision-making for both banks and applicants.
☐ Need for faster credit card approval. ☐ Increasing number of applications. ☐ Requirement for accurate and automated decision-making. ☐ Reducing operational costs and manual effort.	6. CUSTOMER LIMITATIONS [CL] • Limited knowledge of approval criteria. • Time constraints. • Incomplete financial information. • Dependence on manual bank processes.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Online • Flask Web Application • Bank Websites • Mobile Browsers Offline • Bank Branches • Customer Service Centers